Exact truncated moment boundaries for a Gaussian–cubic polynomial functional

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Abstract

We determine positivity of polynomial squares through degree four and realization by positive measures through degree nine for the formal moment sequence with exponential generating series $\exp(vz^2/2 + kz^3/6)$. For $v > 0$, put $r = k^2/v^3$. Quadratic, cubic and quartic square tests give successively

$$r \le 2, \qquad 12 - 24r - 9r^2 \ge 0, \qquad 8 - 40r - 45r^3 \ge 0.$$

The last inequality is exactly the criterion for matching moments through degree eight by a positive probability measure. Matching the prescribed ninth moment requires the strict inequality. At the quartic null boundary a positive measure matches all eight moments, but the null polynomial forces a different ninth moment. A parallel obstruction occurs between degrees six and seven. Finite Jacobi matrices construct the representing measures and steer an arbitrary ninth moment in the strict region. An attained sixth-moment bound gives a concrete low-degree diagnostic. The methods are classical truncated moment theory; the explicit parameter formulas are matched to a frozen Lean development, with no priority claim.

1 The polynomial functional and the main boundaries

Let $v, k \in \mathbb{R}$. Define $m_0 = 1$ and, for $n \ge 0$,

$$m_{n+1} = vn\, m_{n-1} + \frac{k}{2} n(n-1) m_{n-2}, \tag{1}$$

where terms with negative indices are zero. Define the linear functional $\mathcal{L}_{v,k}$ on $\mathbb{R}[x]$ by $\mathcal{L}_{v,k}(x^n) = m_n$. The generating series identity

$$\sum_{n \ge 0} m_n \frac{z^n}{n!} = \exp(vz^2/2 + kz^3/6) \tag{2}$$

is an identity of formal power series: differentiation gives (1) and the constant coefficient is one. It is not the assumption that a positive law has this moment-generating function.

Say that $\mathcal{L}_{v,k}$ passes the degree-d square test when $\mathcal{L}_{v,k}(p^2) \ge 0$ for every real polynomial of degree at most d. A Borel probability measure represents the sequence through degree N when $\int |x|^j \, d\mu < \infty$ and $\int x^j \, d\mu = m_j$ for $0 \le j \le N$. Introduce

$$\begin{aligned} \delta &= 2v^3 - k^2, \\ D &= 12v^6 - 24v^3k^2 - 9k^4, \\ Q &= 288v^{10} - 1440v^7k^2 - 1620vk^6. \end{aligned} \tag{3}$$

Theorem 1.1. *For $v > 0$ the degree-two, degree-three and degree-four square tests hold exactly when $\delta \geq 0$, $D \geq 0$ and $Q \geq 0$, respectively. A positive probability measure represents the moments through degree eight exactly when $Q \geq 0$, and through degree nine exactly when $Q > 0$. All positive constructions may be chosen with finite support. If $Q > 0$, any real ninth moment can be prescribed while keeping the first eight moments fixed.*

The case $v = 0$ is degenerate: a positive measure with the first three prescribed moments must be δ_0 and $k = 0$. If $v < 0$, its second moment already excludes a positive measure. Thus the full degree-eight criterion is $(v, k) = (0, 0)$ or $v > 0, Q \geq 0$; at degree nine replace $Q \geq 0$ by $Q > 0$.

The issue at $Q = 0$ is not failure of the preceding square test. It is compatibility of the next moment with the null polynomial. We give the positive constructions and the incompatibility proof explicitly.

The general tools are classical. Curto and Fialkow's odd truncated Hamburger criterion tests both positivity and a range condition; their even criterion likewise distinguishes singular data [1]. Here the contribution under examination is an explicit evaluation for a two-parameter sequence and a verified exposition of its boundaries. We do not claim a new general moment theorem or established novelty for these formulas.

2 Moment matrices and exact square tests

The recursion gives

j	m_j	j	m_j
0	1	5	$10vk$
1	0	6	$15v^3 + 10k^2$
2	v	7	$105v^2k$
3	k	8	$105v^4 + 280vk^2$
4	$3v^2$	9	$1260v^3k + 280k^3$

(4)

Let $H_d = (m_{i+j})_{0 \leq i,j \leq d}$. The square test is exactly $H_d \succeq 0$, since the coefficient vector of p has quadratic form $\mathcal{L}_{v,k}(p^2)$.

Use the monic centered basis

$$U_0 = 1, \qquad U_1 = x, \qquad U_2 = x^2 - v,$$
$$U_3 = x^3 - 3vx - k, \qquad U_4 = x^4 - 6vx^2 - 4kx + 3v^2.$$

Each U_j, $j \geq 1$, has zero functional mean. This change of basis is triangular with diagonal one. On (U_1, U_2, U_3) the Gram matrix is

$$G_3 = \begin{pmatrix} v & k & 0 \\ k & 2v^2 & 6vk \\ 0 & 6vk & 6v^3 + 9k^2 \end{pmatrix}, \qquad \det G_3 = D. \tag{5}$$

Adjoining U_4 gives

$$G_4 = \begin{pmatrix} v & k & 0 & 0 \\ k & 2v^2 & 6vk & 6k^2 \\ 0 & 6vk & 6v^3 + 9k^2 & 36v^2k \\ 0 & 6k^2 & 36v^2k & 24v^4 + 144vk^2 \end{pmatrix}, \qquad \det G_4 = Q. \tag{6}$$

These identities follow by multiplying the displayed polynomials and using (4). In particular the leading determinants of H_2, H_3, H_4 are δ, D, Q.

Proposition 2.1. *For $v > 0$ the three square-test criteria in Theorem 1.1 hold.*

Proof. For $p = c + bx + ax^2$, direct completion of squares yields

$$v\mathcal{L}_{v,k}(p^2) = v(c + va)^2 + (vb + ka)^2 + \delta a^2.$$

This proves sufficiency of $\delta \geq 0$. If $\delta < 0$, take $(c, b, a) = (-v^2, -k, v)$; its square has value $v\delta < 0$.

Necessity of $D \geq 0$ and $Q \geq 0$ follows because a positive semidefinite matrix has nonnegative determinant. For sufficiency put $r = k^2/v^3 \geq 0$. If $D \geq 0$, then $12 - 24r - 9r^2 \geq 0$, which forces $r < 1/2$ and therefore $\delta > 0$. The leading 2×2 block of G_3 is positive definite. Its scalar Schur complement is $D/\delta \geq 0$, so $G_3 \succeq 0$. If $Q \geq 0$, its factorization

$$Q = 36v^{10}(8 - 40r - 45r^3) \tag{7}$$

implies $r \leq 1/5$. Hence $D/v^6 \geq 12 - 24/5 - 9/25 = 171/25 > 0$. Thus G_3 is positive definite and the last Schur complement in G_4 is $Q/D \geq 0$. This proves $G_4 \succeq 0$. The constant basis vector is orthogonal to these centered blocks and has norm one, completing the argument for H_d. □

This proof also identifies all positive pivots needed in the representing matrices below. To display exact null directions, define

$$R_3 = 6vk^2U_1 - 6v^2kU_2 + \delta U_3, \tag{8}$$

$$\begin{aligned} R_4 = {} & -18k^3(10v^3 - 3k^2)U_1 + 18vk^2(10v^3 - 3k^2)U_2 \\ & - 72v^2k(v^3 - k^2)U_3 + DU_4. \end{aligned} \tag{9}$$

Multiplication by the Gram matrices shows that R_3 is orthogonal to every polynomial of degree at most two, and R_4 to every polynomial of degree at most three. Their square values are

$$\mathcal{L}_{v,k}(R_3^2) = \delta D, \qquad \mathcal{L}_{v,k}(R_4^2) = DQ. \tag{10}$$

When the preceding pivots are positive, the monic orthogonal polynomials have squared norms

$$h_0 = 1, \qquad h_1 = v, \qquad h_2 = \delta/v, \qquad h_3 = D/\delta, \qquad h_4 = Q/D. \tag{11}$$

For example, $P_2 = U_2 - (k/v)U_1$, $P_3 = R_3/\delta$ and $P_4 = R_4/D$. These computations can be checked from the small Gram matrices without expanding large determinants.

3 A finite spectral construction, including a null last pivot

We spell out the finite construction to handle the semidefinite boundary rather than invoke positivity alone. The lemma is standard Jacobi-matrix moment theory.

Lemma 3.1. *Let L be a real linear functional on polynomials of degree at most $2d$, with $L(1) = 1$. Suppose H_{d-1} is positive definite and H_d is positive semidefinite. There is a real symmetric $(d+1) \times (d+1)$ matrix J_f, depending on one real parameter f, for which*

$$\langle e_0, J_f^n e_0 \rangle = L(x^n), \qquad 0 \leq n \leq 2d.$$

Its spectral probability measure at e_0 represents these moments. If h_d is the last monic orthogonal squared norm, then

$$\langle e_0, J_f^{2d+1} e_0 \rangle = \langle e_0, J_0^{2d+1} e_0 \rangle + h_d f. \tag{12}$$

The construction remains valid when $h_d = 0$.

Proof. Give $V_{d-1} = \mathbb{R}[x]_{\le d-1}$ the positive definite inner product $L(pq)$. Let $q_0 = 1, q_1, \dots, q_{d-1}$ be its orthonormal polynomials with positive leading coefficients. Orthogonal projection of x^d onto V_{d-1} defines the monic residual P_d and $h_d = L(P_d^2) \ge 0$. For $p = p_{<d} + cx^d$ in V_d, define a coordinate map into $\mathbb{R}^{d+1}$ by taking the coordinates of $p - cP_d \in V_{d-1}$ and giving it last coordinate $c\sqrt{h_d}$. Orthogonality proves

$$\langle C(p), C(q)\rangle = L(pq), \qquad p, q \in V_d. \tag{13}$$

When $h_d = 0$ this is a map of the semidefinite quotient; no division by h_d is used.

Specify the first d columns of J_f by $J_f C(p) = C(xp)$ for $p \in V_{d-1}$. These columns are symmetric on their first d entries, because $L(xpq) = L(pxq)$. Fill the remaining off-diagonal entries by symmetry, and set the last diagonal to f. This defines a real symmetric matrix, including at $h_d = 0$. By induction $J_f^j e_0 = C(x^j)$ for $0 \le j \le d$: only the first d columns are used until the last step. For $n \le 2d$ choose $i, j \le d$ with $i + j = n$. Then

$$\langle e_0, J_f^n e_0\rangle = \langle J_f^i e_0, J_f^j e_0\rangle = L(x^{i+j})$$

by (13).

Let $J_f = \sum_\lambda \lambda \Pi_\lambda$ be its finite orthogonal spectral decomposition. The measure

$$\mu_f = \sum_\lambda \|\Pi_\lambda e_0\|^2 \delta_\lambda \tag{14}$$

has nonnegative weights summing to one. Its nth moment is exactly $\langle e_0, J_f^n e_0\rangle$, and finite support supplies all absolute moments.

Finally $J_f - J_0 = f e_d e_d^{\mathsf{T}}$, while $J_f^d e_0 = C(x^d)$ is independent of f and has last coordinate $\sqrt{h_d}$. Thus

$$\langle e_0, J_f^{2d+1} e_0\rangle = \langle C(x^d), J_f C(x^d)\rangle = \langle C(x^d), J_0 C(x^d)\rangle + h_d f.$$

This proves (12). □

The symmetric matrices in the lemma are tridiagonal. Indeed if $i < j - 1$, orthogonality gives $L(xq_i q_j) = 0$, because xq_i has degree below j. If P_j is monic with norm $h_j > 0$, the neighboring entry is $\sqrt{h_j/h_{j-1}}$ by its leading coefficient. At $h_d = 0$, the last row and column decouple. The spectral measure at e_0 then lives on the preceding block; an arbitrary last diagonal introduces no mass.

4 Eight moments and a freely chosen ninth moment

Assume $v > 0$ and $Q \ge 0$. By Proposition 2.1, H_3 is positive definite and H_4 positive semidefinite. Lemma 3.1 with $d = 4$ gives a positive probability measure representing all moments through eight. Conversely such a measure makes every degree-four square nonnegative, so $Q \ge 0$. This proves the degree-eight classification, including its boundary.

For direct reproduction, the matrix is

$$J_f = \begin{pmatrix} 0 & b_1 & 0 & 0 & 0 \\ b_1 & a_1 & b_2 & 0 & 0 \\ 0 & b_2 & a_2 & b_3 & 0 \\ 0 & 0 & b_3 & a_3 & b_4 \\ 0 & 0 & 0 & b_4 & f \end{pmatrix}, \tag{15}$$

where

$$b_1 = \sqrt{v}, \qquad b_2 = \sqrt{\delta}/v, \qquad b_3 = \sqrt{vD}/\delta, \qquad b_4 = \sqrt{\delta Q}/D, \tag{16}$$

$$a_1 = k/v, \qquad a_2 = \frac{k(4v^3+k^2)}{v\delta}, \qquad a_3 = \frac{72v^2k(v^3-k^2)}{D} - \frac{6v^2k}{\delta}. \tag{17}$$

All denominators are strictly positive. The off-diagonal entries follow from (11); the diagonal entries are $\mathcal{L}_{v,k}(xP_j^2)/h_j$ for $j = 1, 2, 3$. For example $\mathcal{L}_{v,k}(xP_1^2) = k$, and inserting $P_2 = x^2 - (k/v)x - v$ gives $\mathcal{L}_{v,k}(xP_2^2) = 4vk + k^3/v^2$, which yields a_2. The same substitution of R_3/δ gives a_3. Thus (15) is the concrete form of the construction, not a matrix whose moment matching is assumed.

Write $s_9(f) = \int x^9 \, d\mu_f$. The steering formula is

$$s_9(f) = s_9(0) + (Q/D)f. \tag{18}$$

If $Q > 0$, any target $s \in \mathbb{R}$ is obtained by $f = (s - s_9(0))D/Q$. In particular one can choose $s = m_9 = 1260v^3k + 280k^3$. This proves the positive degree-nine construction and the arbitrary-target assertion. At $Q = 0$, however, (18) has zero slope. The next section shows that the prescribed m_9 is incompatible with every positive representation, not merely with this chosen family of matrices.

5 The null boundary excludes the prescribed ninth moment

We first record the general compatibility argument.

Lemma 5.1. *Suppose a positive measure represents L through degree $2d+1$, and p has degree at most d with $L(p^2) = 0$. Then $L(xp^2) = 0$.*

Proof. Polynomial integrability follows from the assumed absolute moments. Since $p^2 \geq 0$ and its integral is zero, $p = 0$ almost everywhere. Therefore $xp^2 = 0$ almost everywhere, and its integral is zero. □

For the particular polynomial R_4 of (9), direct multiplication with the moments through degree nine gives the exact identity

$$\begin{aligned} v\mathcal{L}_{v,k}(xR_4^2) = {} & (-9k^5 + 48v^3k^3 - 108v^6k)Q \\ & + 864v^{10}k(52v^6 - 272v^3k^2 + 213k^4). \end{aligned} \tag{19}$$

This is a polynomial identity in v, k, verifiable from the finite table; the accompanying symbolic checker independently verifies its transcription.

Now assume $v > 0, Q = 0$. Its normalized equation is $8 - 40r - 45r^3 = 0$, where $r = k^2/v^3 \geq 0$. It implies $r \leq 1/5$, and $k \neq 0$ since the constant term is eight. On $0 \leq r \leq 1/5$ the polynomial $52 - 272r + 213r^2$ is decreasing, as its derivative $-272 + 426r$ is negative. Its minimum on this interval is $153/25 > 0$. Hence the second line of (19) is nonzero, and $\mathcal{L}_{v,k}(xR_4^2) \neq 0$. But (10) gives $\mathcal{L}_{v,k}(R_4^2) = 0$. Lemma 5.1 rules out a positive degree-nine representation. Together with the preceding section, this completes Theorem 1.1 for $v > 0$.

The degenerate cases stated after the theorem also follow without a moment-extension theorem. If $v = 0$, a positive representing measure has $x = 0$ almost everywhere, so $k = 0$ and it must be δ_0. Conversely (1) gives $m_j = 0$ for every $j > 0$ when $(v, k) = (0, 0)$. Negative v is impossible because $\int x^2 \, d\mu \geq 0$.

The boundary is nonempty. The polynomial $g(r) = 8 - 40r - 45r^3$ is strictly decreasing for $r \geq 0$, with $g(0) > 0$ and $g(1/5) < 0$. It has a unique positive root r_4, and $g(1/6) = 9/8 > 0$ shows

$$1/6 < r_4 < 1/5. \tag{20}$$

For $v = 1$, $k = \sqrt{r_4}$ supplies an explicit algebraic parameter at which all eight moments are realizable but the prescribed ninth is not.

6 The preceding six-to-seven obstruction

The smaller null boundary illustrates the same mechanism at a different truncation and is part of the same story.

Proposition 6.1. *If $v > 0$ and $D = 0$, a positive probability measure represents the moments through degree six, but none represents them through degree seven.*

Proof. The equation $D = 0$ gives $12 - 24r - 9r^2 = 0$, with

$$r = r_3 = (2\sqrt{7} - 4)/3 < 1/2. \tag{21}$$

Thus $\delta > 0$, H_2 is positive definite and H_3 is positive semidefinite. Lemma 3.1 with $d = 3$ constructs the degree-six representation. The last pivot $h_3 = D/\delta$ is zero, so it suffices to retain the preceding three-dimensional block.

The null cubic R_3 satisfies $\mathcal{L}_{v,k}(R_3^2) = \delta D = 0$. Its next cross moment is

$$\mathcal{L}_{v,k}(xR_3^2) = -14v^2kD + 24v^5k(10v^3 - 17k^2). \tag{22}$$

Again this follows by polynomial multiplication and (4). At $D = 0$ one has $k \neq 0$ and $10v^3 - 17k^2 = v^3(10 - 17r_3) > 3v^3/2 > 0$. So (22) is nonzero, contradicting Lemma 5.1 for any degree-seven representation. □

There is also an open gap between the cubic and quartic square tests. At $v = 1, k = 1/2$, one has $r = 1/4$ and

$$D = 87/16 > 0, \qquad Q/36 = 8 - 40/4 - 45/64 = -173/64 < 0.$$

Continuity preserves both strict signs on a neighborhood. Thus even strict cubic square positivity does not guarantee quartic square positivity or eight-moment realizability. This example is a direct specialization of the exact criteria, not an appeal to numerical eigenvalues.

7 An attained sixth-moment bound

The formal sequence can fail much earlier at particular parameters. A sharp bound explains the obstruction at $v = k = 1$ and supplies an independent finite construction.

Proposition 7.1. *If a probability measure has moments*

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = 1, \quad m_3 = 1, \quad m_4 = 3, \quad m_5 = t,$$

and finite sixth absolute moment, then

$$m_6 \geq t^2 - 8t + 26. \tag{23}$$

For every real t equality is attained by a finitely supported probability measure.

Proof. Put $p_t(x) = x^3 + (4-t)x^2 + (t-7)x + (t-5)$. Expanding its square with the prescribed first five moments gives

$$\int p_t(x)^2 \, d\mu = m_6 - (t^2 - 8t + 26),$$

which proves the lower bound. For attainment, take the spectral measure at e_0 of

$$A_t = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & t-5 \end{pmatrix}.$$

Its first four coordinate-power vectors are

$$e_0, \quad A_t e_0 = (0,1,0), \quad A_t^2 e_0 = (1,1,1), \quad A_t^3 e_0 = (1,3,t-4).$$

Their inner products give moments $m_0, \ldots, m_4$ as stated, and

$$\langle A_t^2 e_0, A_t^3 e_0 \rangle = t, \qquad \|A_t^3 e_0\|^2 = 1 + 9 + (t-4)^2 = t^2 - 8t + 26.$$

The spectral measure is a probability measure with at most three support points and all absolute moments. Its square integral of p_t is zero, so its support is contained in the real zero set of p_t. □

For the prescribed sequence at $v = k = 1$, $m_5 = 10$ and $m_6 = 25$, whereas (23) requires $m_6 \geq 46$. Indeed $p_{10} = x^3 - 6x^2 + 3x + 5$ satisfies $\mathcal{L}_{1,1}(p_{10}^2) = -21$. The degree-two square test still passes, since $k^2 = 1 \leq 2v^3$. This contrasts a finite positive truncation with the next genuinely incompatible data.

8 Normalization, interpretation and exact formal scope

The scale $r = k^2/v^3$ is intrinsic for $v > 0$. Replacing x by $\sqrt{v}\, y$ transforms the formal series to $\exp(z^2/2 + \kappa z^3/6)$ with $\kappa = k/v^{3/2}$. Changing k to $-k$ is reflection of the real variable. Thus the displayed boundaries depend only on $r = \kappa^2$. The hierarchy is

$$0 < r_4 < 1/5 < r_3 < 1/2 < 2,$$

where r_4 is the unique positive root of $8 - 40r - 45r^3$ and $r_3 = (2\sqrt{7} - 4)/3$. The strict gaps show why matching a few moments cannot justify interpreting the whole functional as expectation under a positive law. Our assertions stop at the stated finite truncations; no all-degree positivity classification is needed or supplied by this paper.

The proof uses elementary Gram matrices, Schur complements, finite spectral measures and null-polynomial compatibility. These are standard truncated Hamburger tools, including the distinction between positivity and extendibility at singular data in Curto–Fialkow [1]. Their odd-case range condition is reflected concretely in (22) and (19); the strict case admits the free last diagonal in (18). A targeted preliminary search did not settle priority for the explicit two-parameter boundaries. Novelty remains unresolved.

The frozen verified baseline is

`bab688bb6140cd6316c80680095a914ea0d346db.`

The exact final declarations are in the namespaces

`MixedCubicFourthDegree`, `MixedCubicEighthRealization`, `MixedCubicNinthRealization`, `MixedCubicQuarticBoundary`, `MixedCubicBoundaryQuartic`, `MixedCubicSharpSixthMoment`,

with prefix `OperatorDerivativeRewrite.Foundation`. The theorem map records declaration lines and frozen source hashes. The general finite spectral lemma is a human reorganization of the explicitly verified Jacobi matrices and steering identities; it is not represented as a newly named formal theorem. The rational example and the normalization discussion are direct deductions, identified separately from the mapped source statements.

References

[1] R. E. Curto and L. A. Fialkow, Recursiveness, positivity, and truncated moment problems, *Houston J. Math.* **17** (1991), no. 4, 603–635. In particular Theorems 3.1 and 3.9. `https://www.math.uh.edu/~hjm/v017n4/0603CURTO.pdf`.